Predicting ICU Stay Length for Acute Traumatic Cervical Spinal Cord Injuries using MIMIC-IV

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1. Non-Technical Executive Summary

This project explores whether we can accurately predict how long someone with an acute traumatic cervical spinal cord injury will stay in the Intensive Care Unit (ICU). Such patients often face severe complications, including respiratory distress, and must remain in the ICU for extended periods. Traditional tools like APACHE-IV sometimes have trouble forecasting the ICU length of stay (LOS) for these specific injuries. We asked if analyzing a patient's first 24 hours of ICU data could improve predictions.

To investigate this, we used the MIMIC-IV database and identified patients with acute traumatic cervical spinal cord injuries (ICD-9 codes 806.0, 806.00–806.09). We found only 97 ICU stays that matched our search due to our narrow focus. We gathered demographic, admission, and clinical information (such as lab tests, vital signs, and medication counts) from the first day of ICU admission. Since LOS was skewed, we applied a log transformation to help stabilize the data. We then tested three models: Lasso Regression, Random Forest, and XGBoost.

The **Random Forest** model gave the most accurate predictions on our small test set, with an average error of about **2.2 days**. It depended strongly on the number of medications and fluids administered in the first 24 hours. Because we only had 97 data points, it remains to be seen if these results hold up in larger studies or against tools like APACHE-IV. Nonetheless, this provides a useful starting point for more focused prediction methods.

2. Technical Exposition

2.1. Introduction and Background

The main question we want to answer is: can a focused machine learning model provide better predictions of ICU LOS for patients with traumatic cervical spinal cord injuries than generic scoring systems like APACHE-IV? Neck-level spinal cord injuries can lead to complications such as respiratory failure, placing substantial demand on ICU resources. Because ICU stays are both expensive and limited, more accurate LOS estimates could help hospitals allocate these resources more effectively. Existing clinical tools often struggle with specialized populations like those with spinal cord injuries, which motivates our data-focused approach.

2.2. Data Wrangling and Cleaning

2.2.1 Data Source

We used MIMIC-IV (version 2.2), a comprehensive database of patient information from Beth Israel Deaconess Medical Center. It contains patient records, admissions, ICU stays, diagnoses, and event-level details like medications and procedures. We processed smaller tables in `pandas` and relied on `dask` for large ones.

2.2.2 Cohort Definition

From `diagnoses_icd.csv.gz`, we extracted all admissions that had ICD-9 codes 806.0, 806.00–806.09. We merged these admissions with the ICU stays table and filtered out non-positive LOS to get **97 ICU stays** for acute traumatic cervical spinal cord injuries. Appendix Figures 9, 11, 16, and 15 give an overview of the cohort's demographics.

2.2.3 Processing Steps

- Converted key time columns (e.g., `admittime`, `intime`) to proper datetime formats.
- Recalculated LOS from `outtime` minus `intime` and excluded invalid or negative values.
- Dropped columns that were mostly empty (over 50% missing) or not relevant to our analysis.
- Merged tables (patients to admissions to icustays) using identifiers, focusing on codes for cervical spinal injuries. Appendix Figures 12 and 13 summarize admission and discharge information.

2.3. Feature Engineering

We concentrated on a 24-hour window plus 6 hours before ICU admission:

- Computed **min, max, mean** for vitals such as heart rate, blood pressure, and respiratory rate.
- Counted medication or fluid inputs, output events, and procedures to assess overall treatment intensity. These distributions appear in Appendix Figures 1, 2, and 3.
- Excluded highly missing features like SpO2 or Bilirubin. Kept a final dataset of aggregated features for modeling.

2.4. Exploratory Data Analysis (EDA)

- **LOS Skewness:** The ICU LOS distribution was strongly right-skewed (Appendix Figure 24 and Q-Q plot in Figure 25), so we used a log transform (`np.log1p`).
- **Multicollinearity:** Some measurements had highly correlated min, max, and mean values (Appendix Figure 8). We either kept only the most informative versions or used models that can handle correlated features.
- **Potential Predictors:** A high medication-fluid count (`count_inputevents_total`) and GCS motor score appeared most relevant (Appendix Figures 4 and 7). Other features showed weaker patterns, for example mean heart rate (Figure 5) or maximum creatinine (Figure 6).

3. Statistical Analysis and Hypothesis Testing

Aside from the initial data exploration, we ran:

- *Shapiro–Wilk test*: Confirmed that LOS is not normally distributed ($p < 0.05$).
- *Spearman correlation*: Showed that total input events and GCS motor score strongly link to LOS, matching EDA plots.
- *Welch's t-test*: Found no significant difference in LOS between male and female groups.
- *One-way ANOVA*: No significant difference in LOS across age groups, though sample sizes for age brackets were small.

4. Modeling Approach

4.1. Data Split and Preprocessing

We split the data into 80% training and 20% testing, ensuring the test portion was held back until final evaluation. Within the training set:

- Numerical features were imputed by median and then standardized.
- Categorical features were imputed by mode and one-hot encoded.

Since LOS was skewed, we used a log transform for model training and took the exponential of predictions to return to the original scale for final metrics.

4.2. Models Tested

In light of our question about more accurate ICU LOS predictions, we chose three models often used for regression tasks on tabular data:

- **Lasso Regression**: A linear model with an L1 penalty that can handle correlated predictors by driving some coefficients to zero. This is a good baseline for seeing if a simpler, regularized approach can capture relationships.
- **Random Forest**: An ensemble of decision trees that can capture non-linearities and interactions. We limited the depth of trees and minimum samples per leaf to prevent overfitting. This model is appealing because it does not assume a linear relationship and often performs well with moderate sample sizes.
- **XGBoost**: A popular gradient boosting technique known for robust performance on many tabular datasets. Like Random Forest, it can handle non-linearities, but it builds trees sequentially and includes built-in regularization. We kept the model relatively shallow to reduce the chance of overfitting our small dataset.

Each model was evaluated by 5-fold cross-validation on the log-transformed LOS. We tracked R^2, RMSE, and MAE on the training splits. Finally, the best model hyperparameters were used to retrain on the entire training set, and predictions were evaluated on the 20% test set in days.

5. Results and Discussion

5.1. Cross-Validation and Test Set Performance

Random Forest and XGBoost outperformed Lasso in cross-validation (Mean CV R^2 on the log scale around 0.845 for Random Forest, 0.802 for XGBoost, and 0.174 for Lasso). Table 1 shows test set results after reversing the log transform:

Model	Test MAE (days)	Test RMSE (days)
Lasso	≈ 2.89	≈ 4.59
RandomForest	$\approx$ **2.21**	$\approx$ **3.84**
XGBoost	≈ 2.57	≈ 4.65

Table 1: Model Performance on the Test Set (Original LOS Scale)

The Random Forest had the lowest MAE (2.2 days), followed by XGBoost. Actual vs. predicted plots (Appendix Figure 20) confirm that Random Forest predictions were closest to the diagonal. Feature importance analysis (Appendix Figure 21) highlights `count_inputevents_total` as the top factor.

5.2. Comparing to APACHE-IV

Our study does not provide a head-to-head comparison against APACHE-IV since:

- Most published APACHE-IV results use broader ICU populations and report R^2 on untransformed LOS, which is not directly comparable to our log-scale analysis in a small subset.
- We did not apply APACHE-IV to these same 97 patients, so we cannot claim we would outperform it.

A fair test would apply APACHE-IV to the exact same cohort and measure MAE in days for a direct comparison.

6. Limitations

- **Small Sample (n=97)**: Our models may overfit, and statistical tests have low power. Conclusions from such a sample size must be interpreted carefully.

- **Single-Center Data**: MIMIC-IV provides data from one hospital system, limiting generalizability.
- **Dropped Features**: Some potentially important data (e.g., SpO2, Bilirubin) were discarded due to high missingness.
- **No Formal APACHE-IV Benchmark**: We did not apply APACHE-IV to these exact patients, so performance comparisons are speculative.
- **Narrow Focus**: We only looked at the first 24 hours (plus 6 hours), and there could be other time-dependent or advanced features that might improve predictions if more data were available.

7. Conclusion and Future Directions

Our Random Forest model achieved a test MAE of about **2.2 days** for acute traumatic cervical spinal cord injury ICU stays, with the total count of medications or fluids administered emerging as the most significant predictor. Although promising, these results come from a dataset of only 97 stays and may not fully generalize.

To build on this work, we recommend:

- Combining MIMIC-IV with other hospital datasets or MIMIC-III to expand the sample size and reduce overfitting risks.
- Running APACHE-IV (or other standard scoring systems) on this same group of 97 patients to directly compare metrics like MAE or RMSE in days.
- Exploring more advanced features or time-series methods, along with better imputation strategies, if more data becomes available.
- Validating these models externally to confirm they hold up beyond a single hospital system.

References

[1] Johnson, A., Bulgarelli, L., Pollard, T., Horng, S., Celi, L. A., & Mark, R. (2023). *MIMIC-IV (version 2.2)*. PhysioNet. https://doi.org/10.13026/6mm1-ek67.

Appendix: Figures

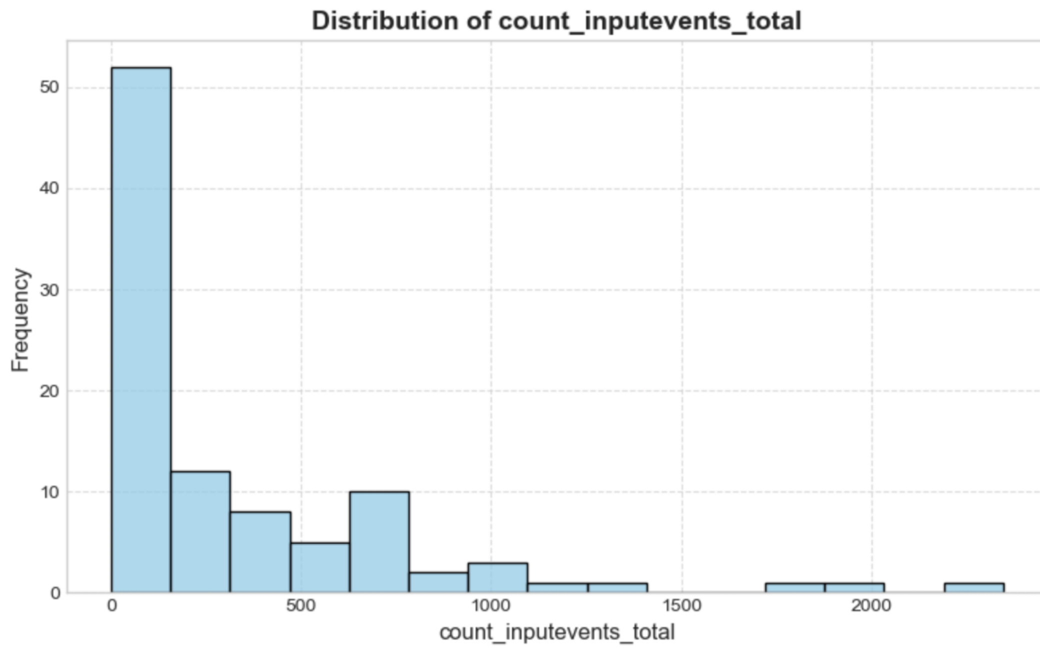


Figure 1: Distribution of `count_inputevents_total`.

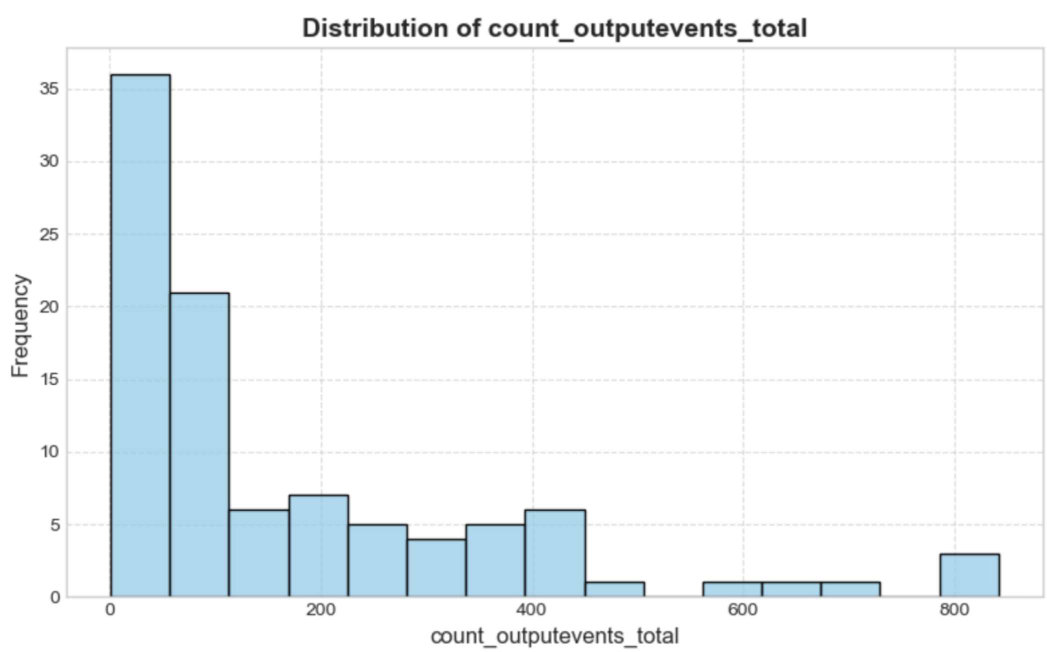


Figure 2: Distribution of `count_outputevents_total`.

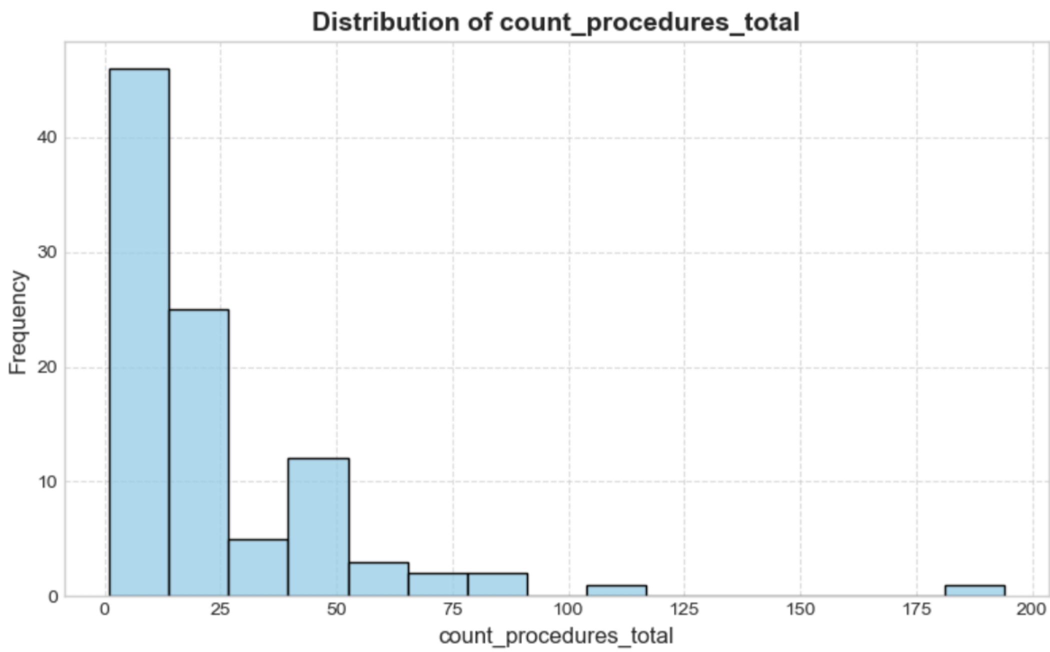


Figure 3: Distribution of `count_procedures_total`.

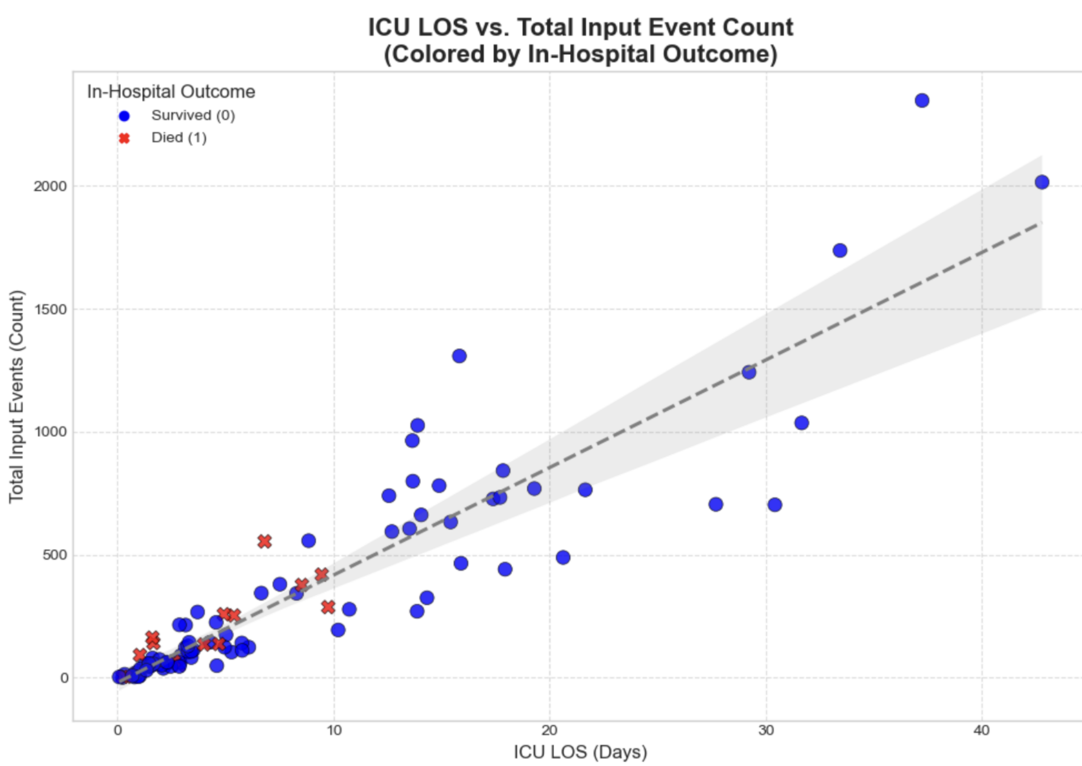


Figure 4: ICU LOS vs. `count_inputevents_total` (first 24h).

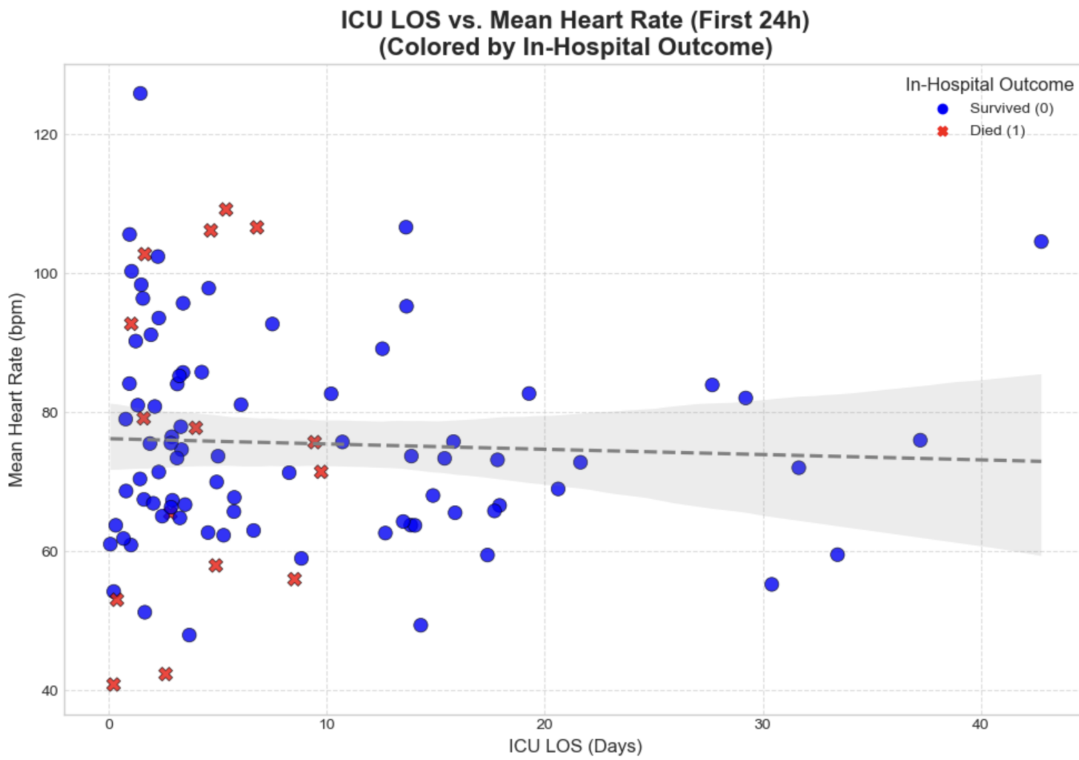


Figure 5: ICU LOS vs. Mean Heart Rate (first 24h).

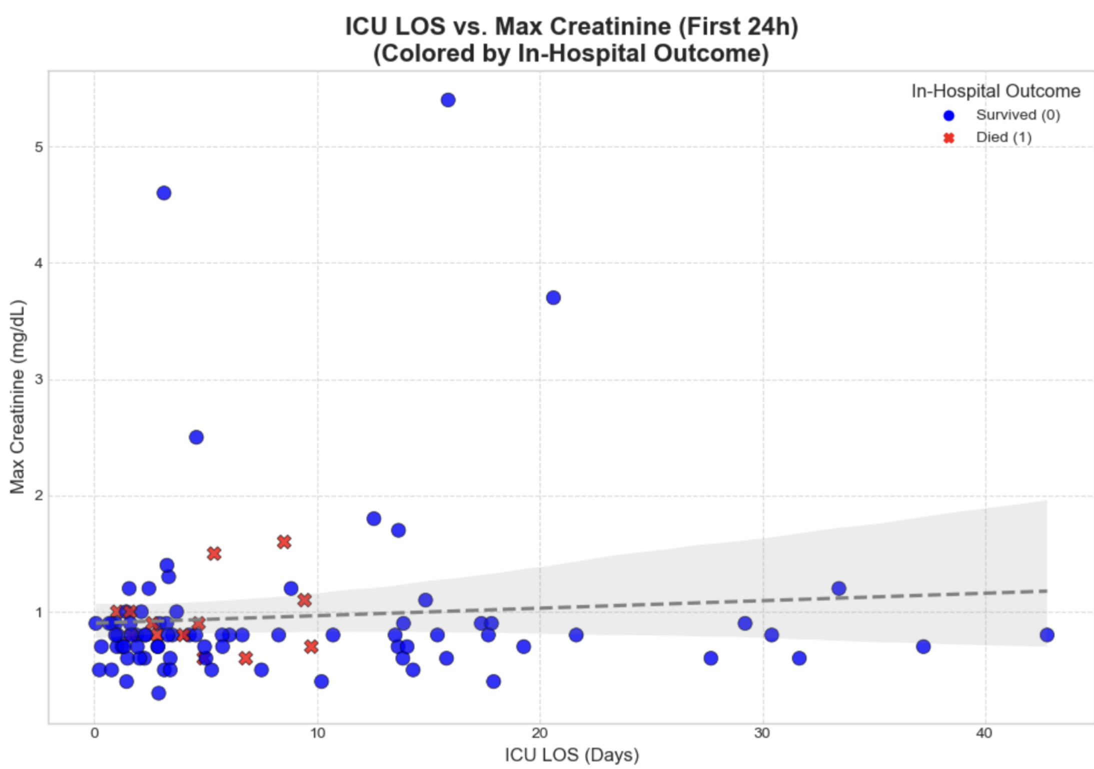


Figure 6: ICU LOS vs. Max Creatinine (first 24h).

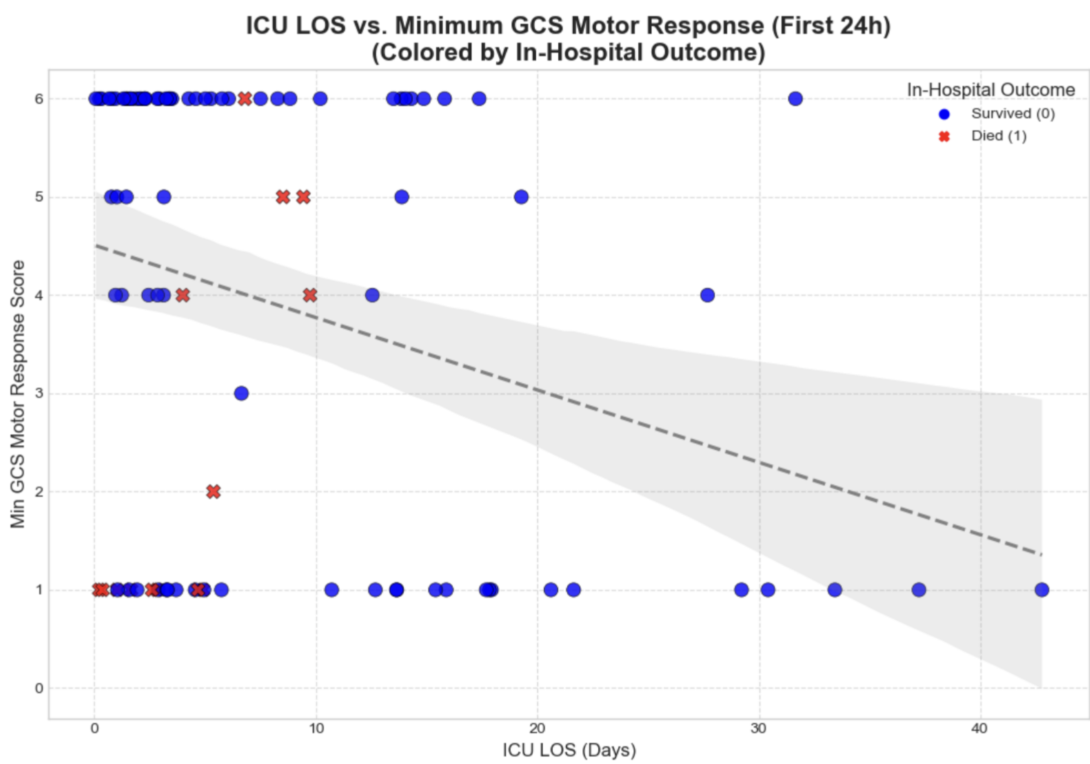


Figure 7: ICU LOS vs. Min GCS Motor (first 24h).

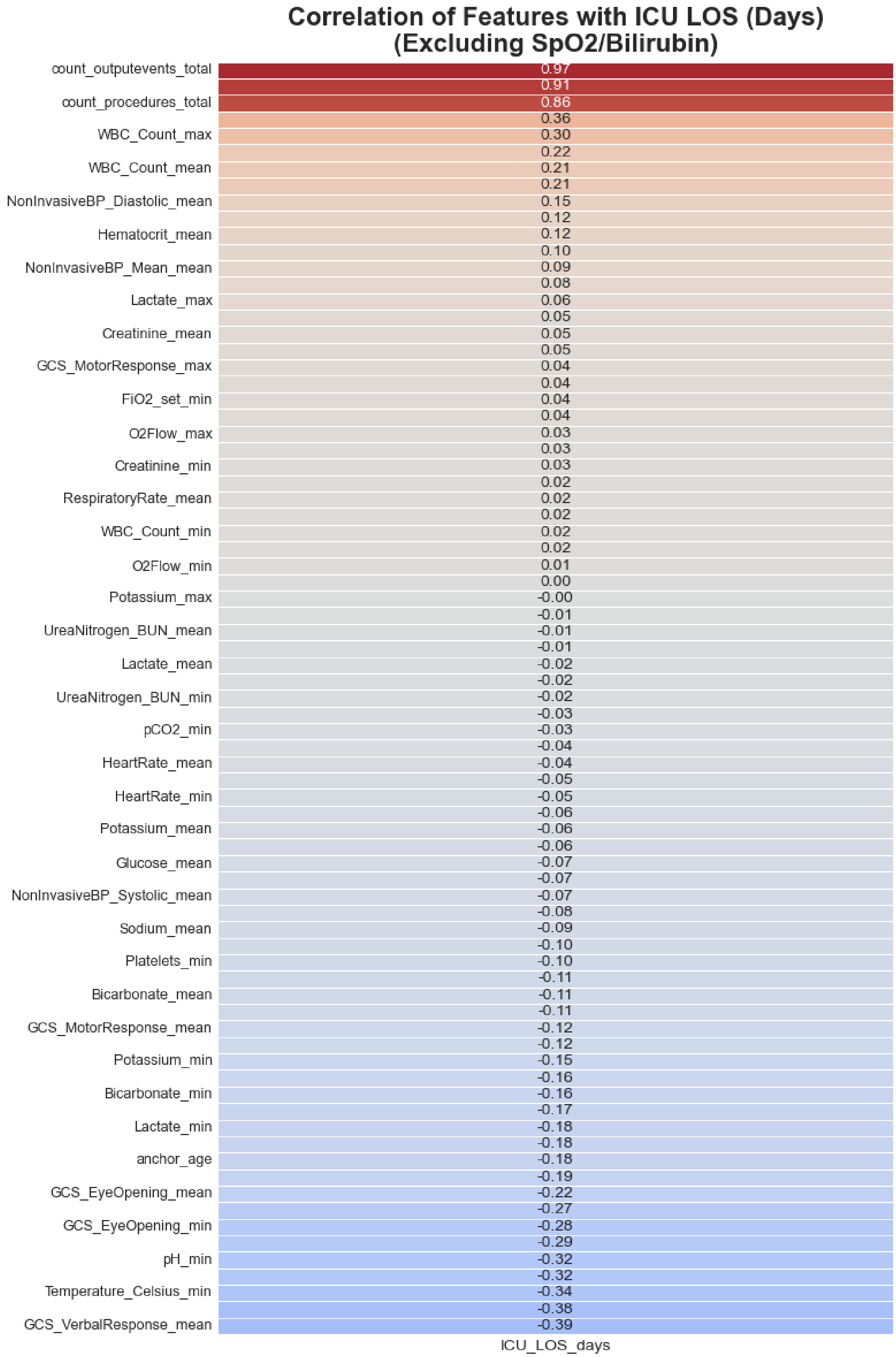


Figure 8: Correlation Heatmap for Key Features.

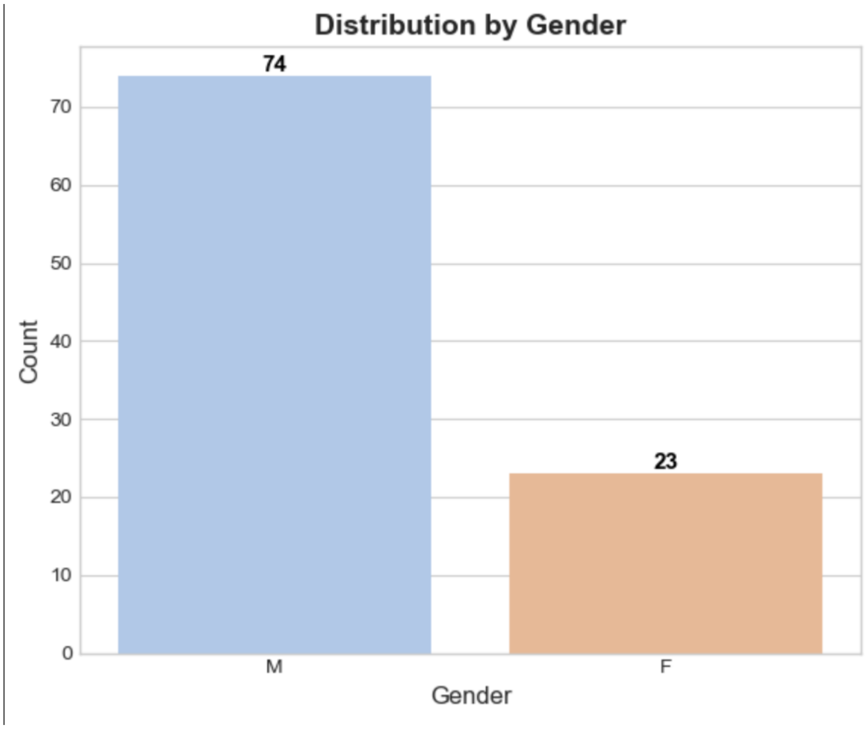
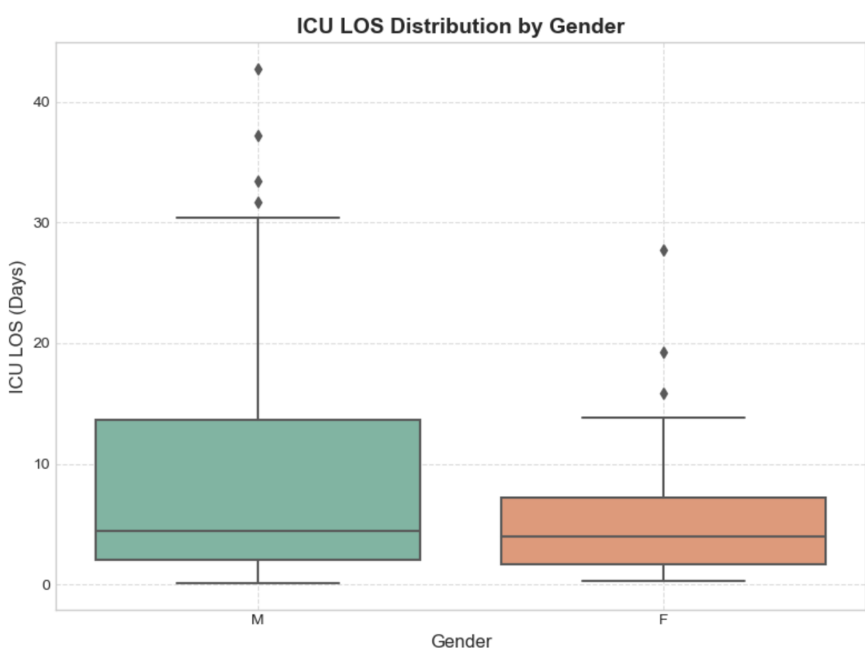


Figure 9: Gender Distribution in the Cohort.



T-test (ICU LOS vs Gender): Statistic=-1.18, p-value=0.243

Figure 10: ICU LOS Distribution by Gender.

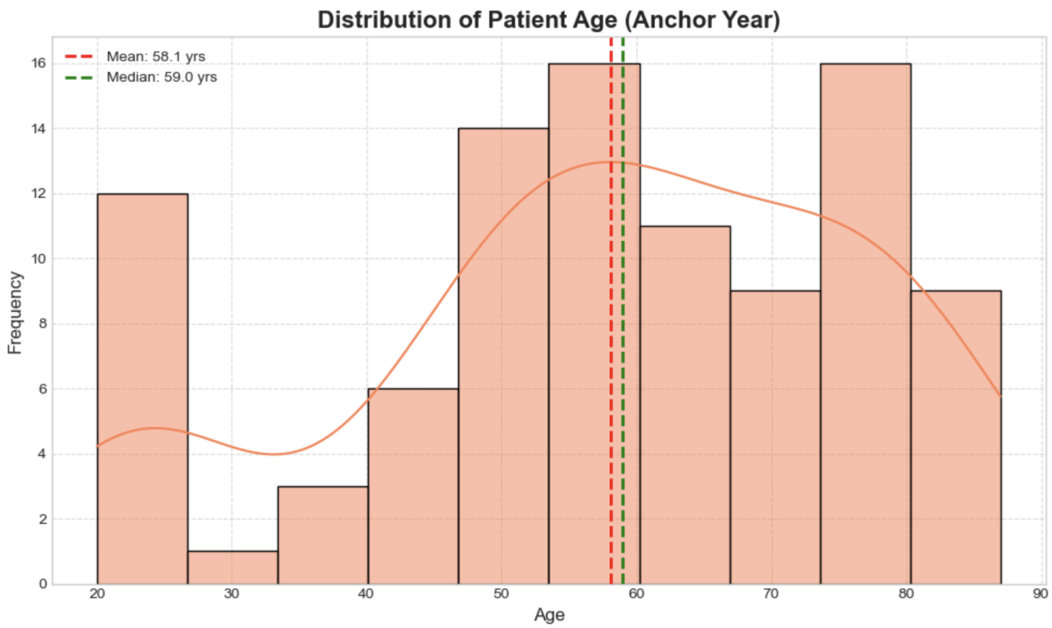


Figure 11: Distribution of Patient Age.

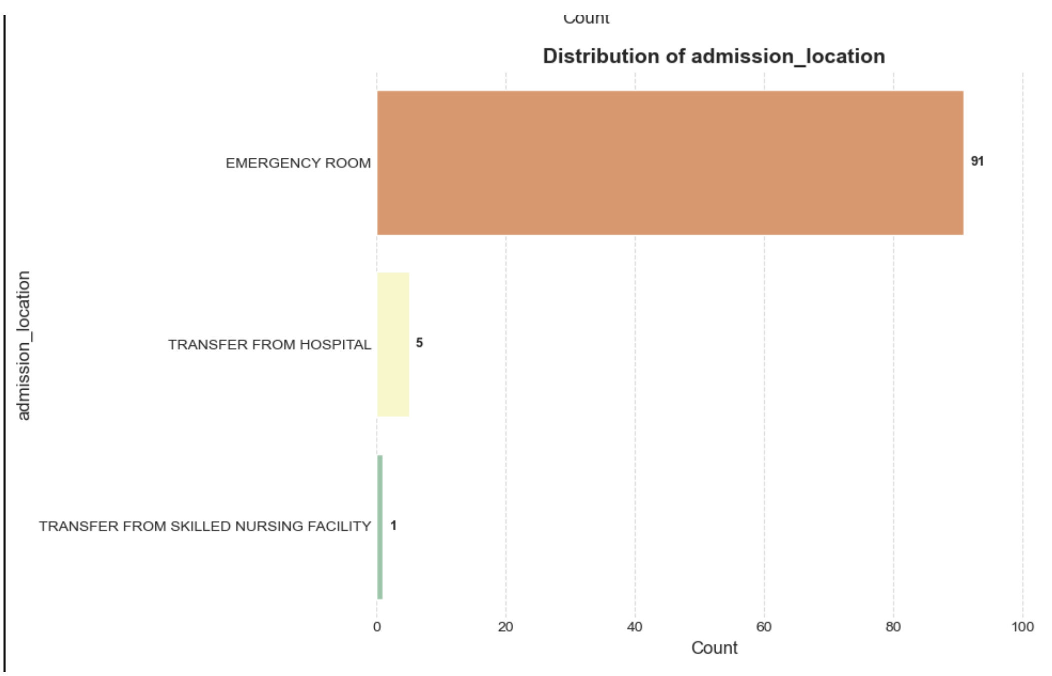


Figure 12: Distribution of Admission Location.

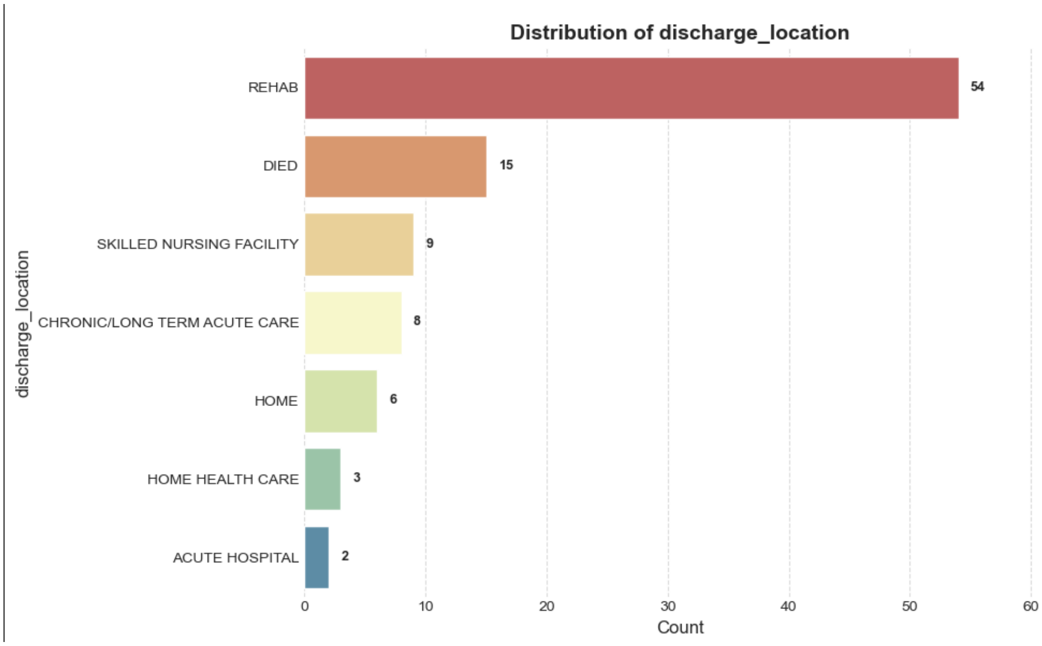


Figure 13: Distribution of Discharge Location.

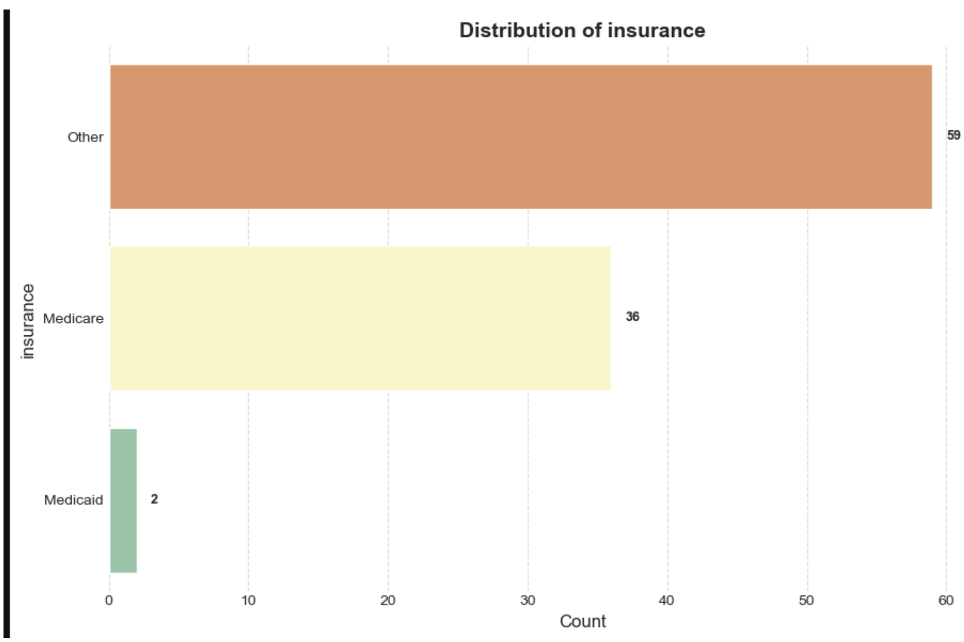


Figure 14: Distribution of Insurance Type.

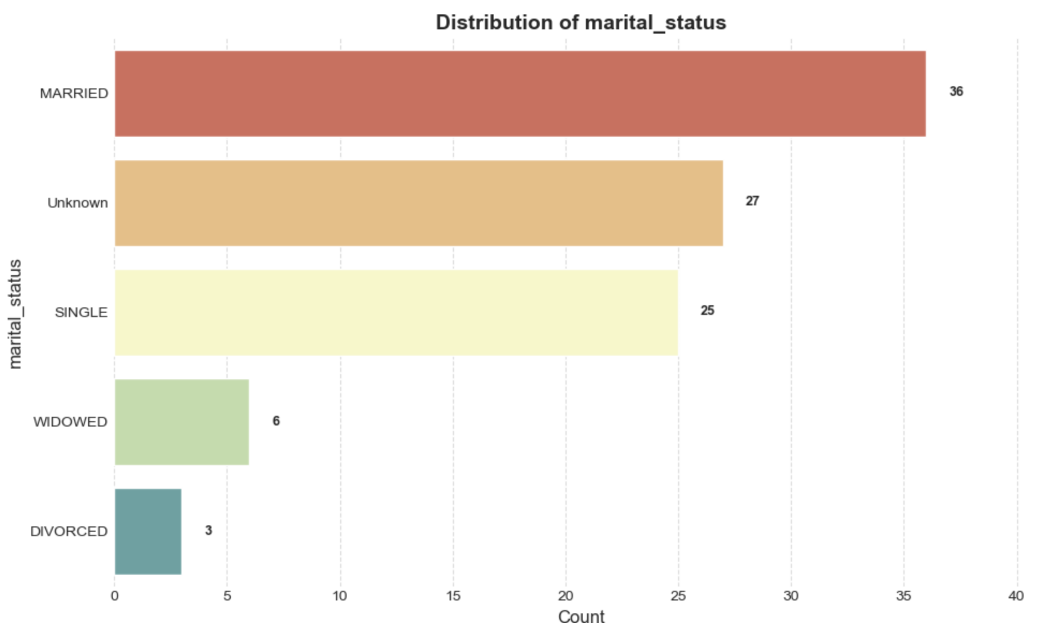


Figure 15: Distribution of Marital Status.

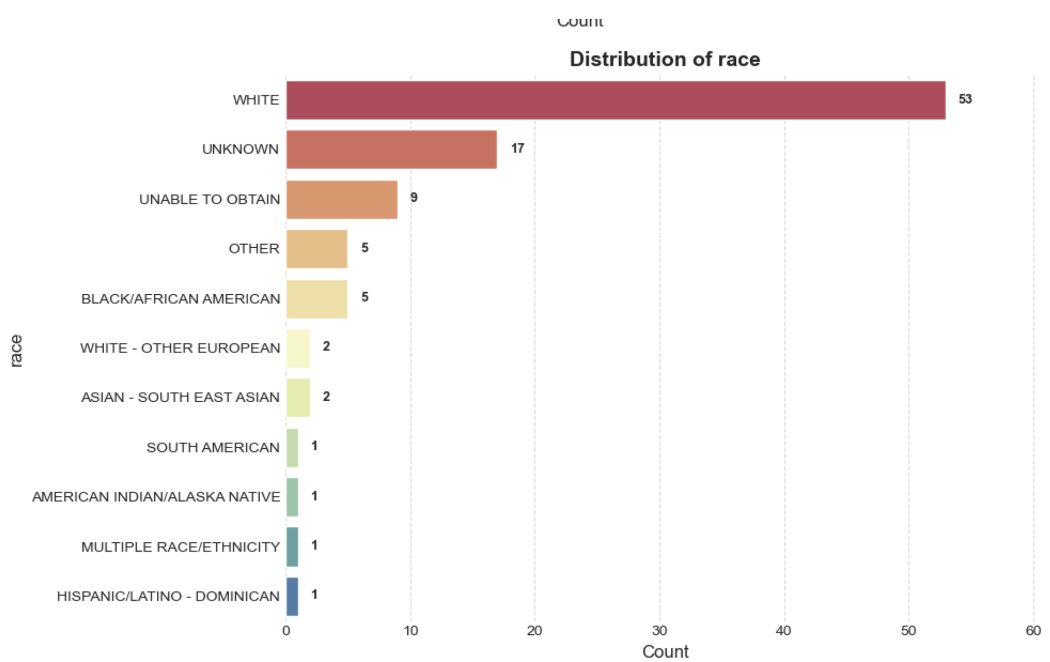


Figure 16: Distribution of Race.

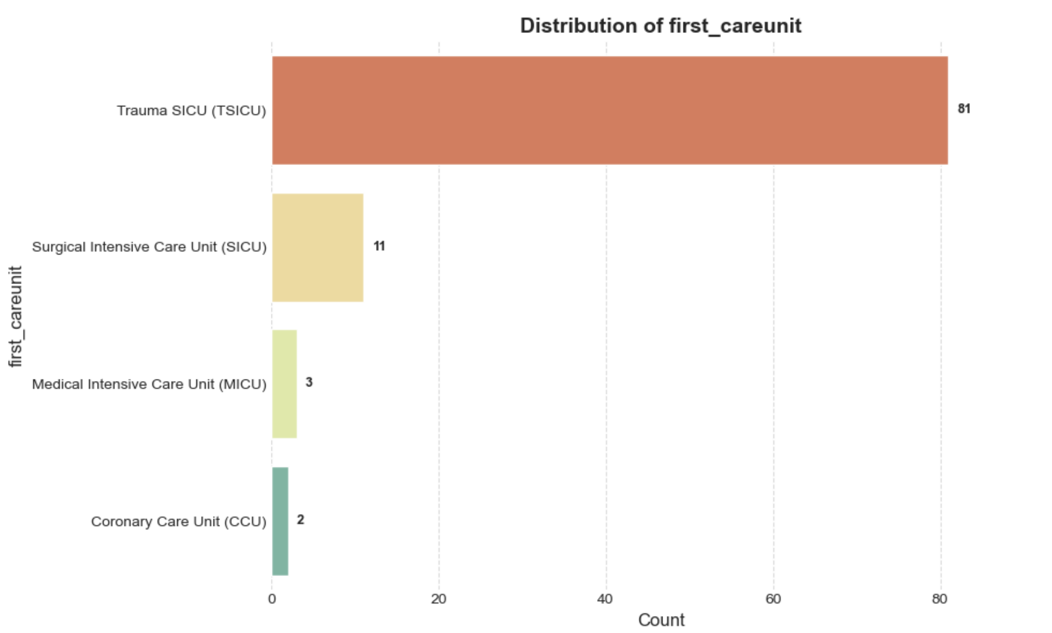


Figure 17: Distribution of First Care Unit.

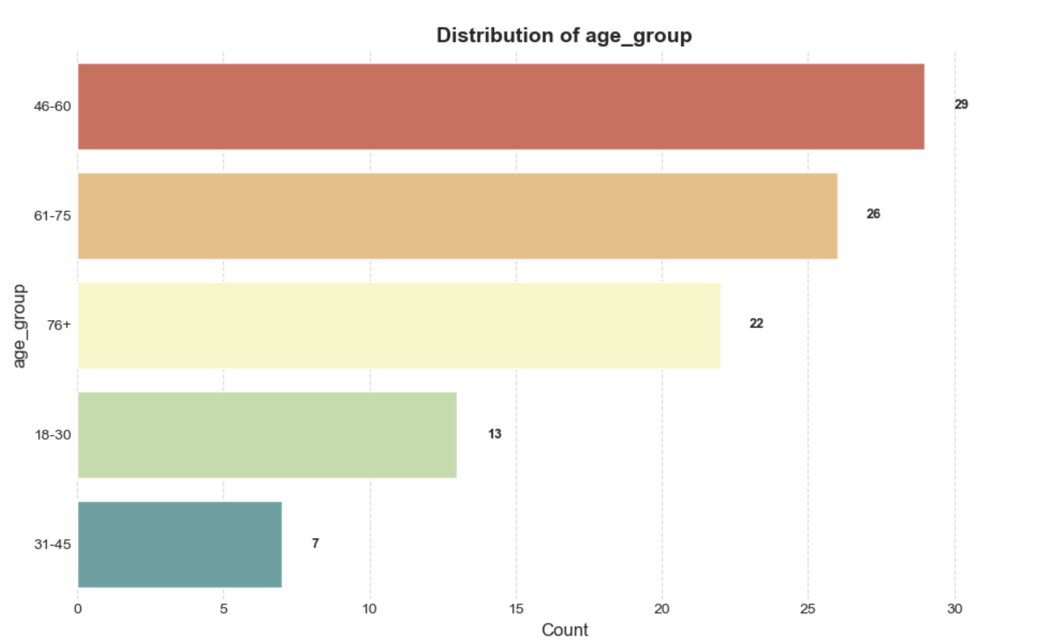


Figure 18: Distribution of Age Group.

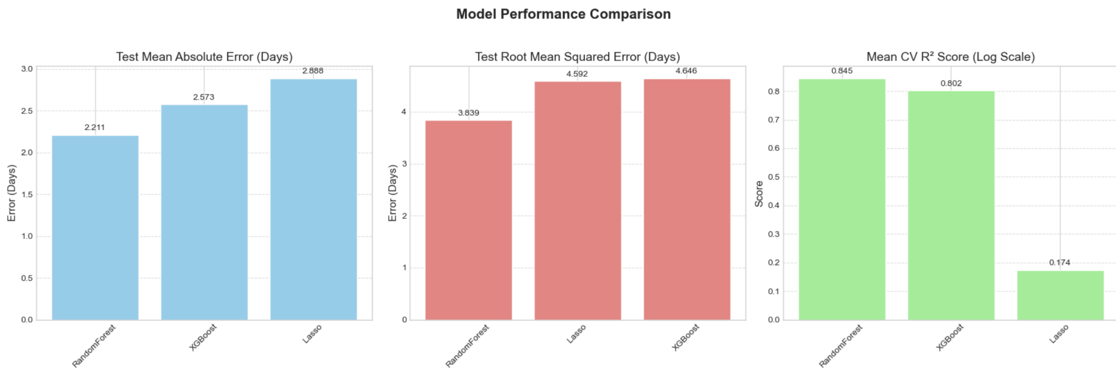


Figure 19: Model Performance Comparison (MAE, RMSE, R^2).

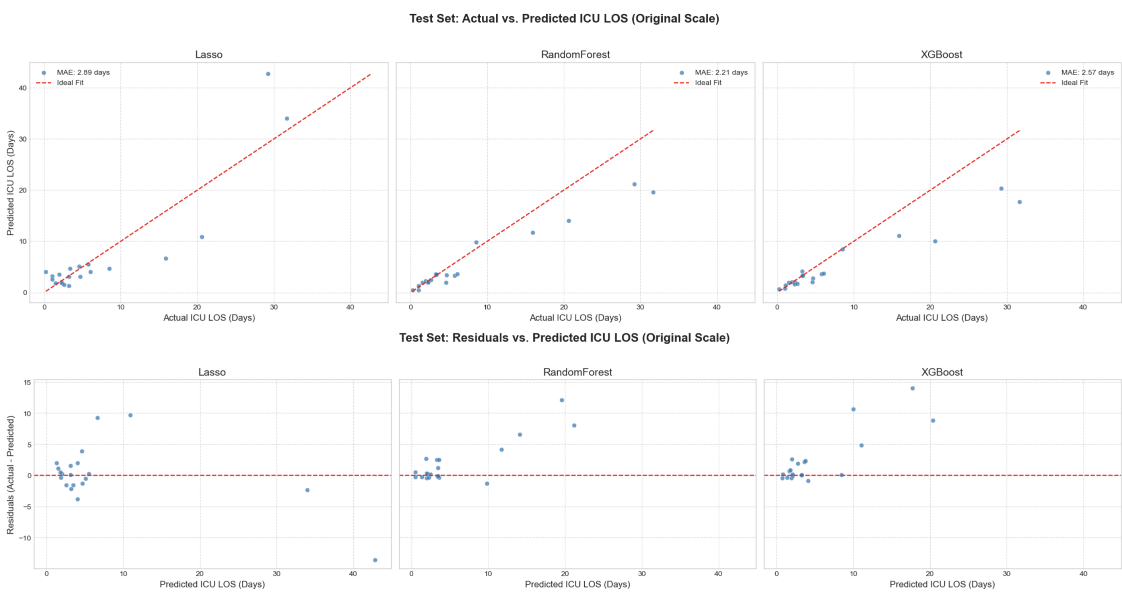


Figure 20: Actual vs. Predicted ICU LOS (Test Set).

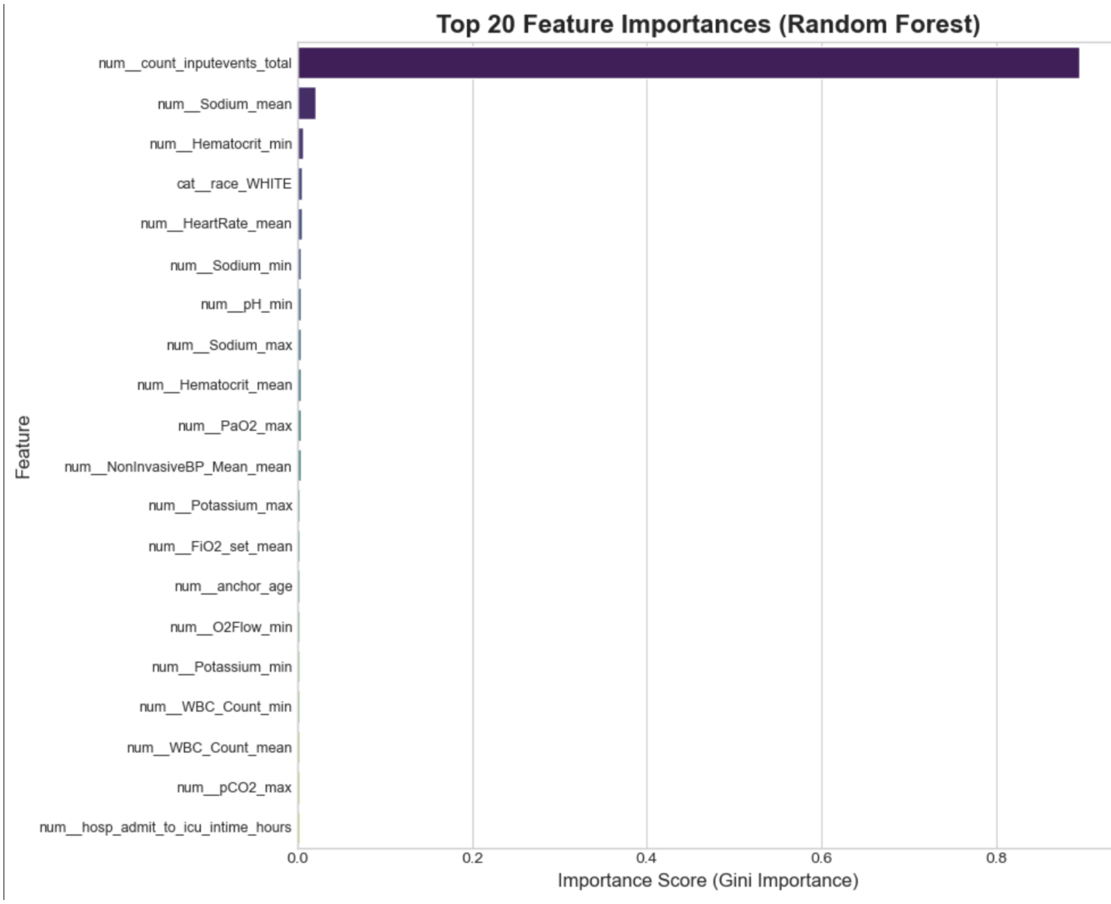


Figure 21: Top 20 Feature Importances (Random Forest).

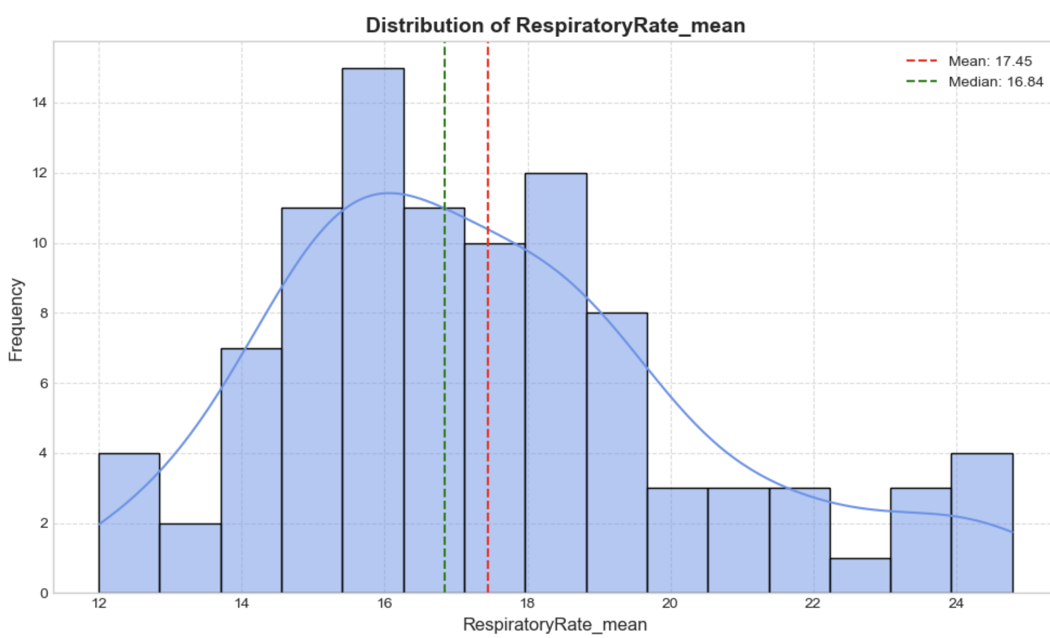


Figure 22: Distribution of Respiratory Rate Mean (First 24h).

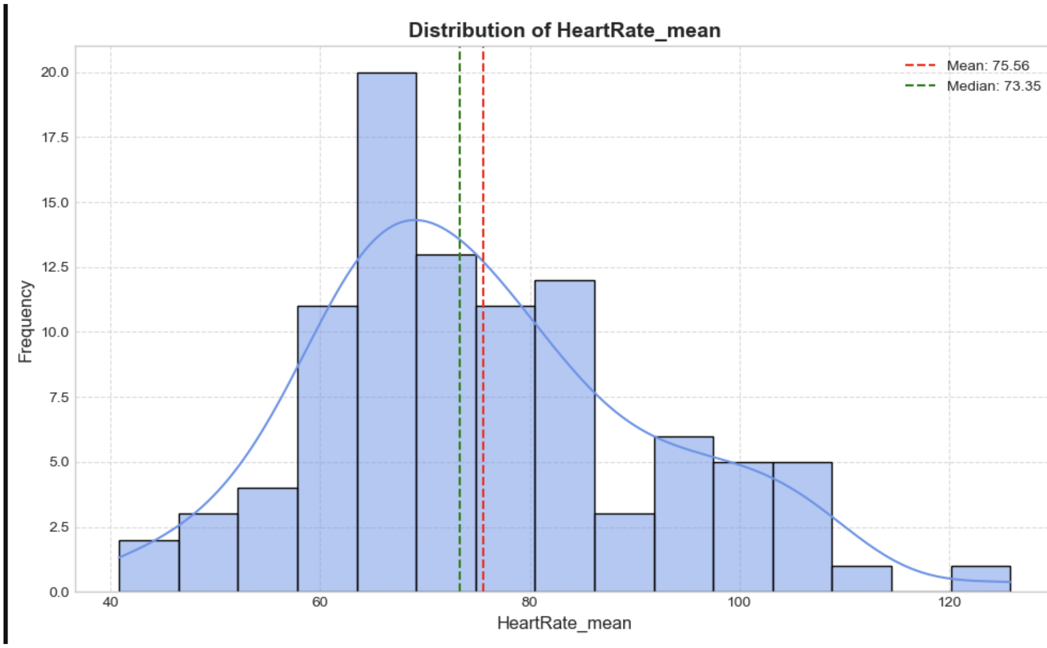


Figure 23: Distribution of Heart Rate Mean (First 24h).

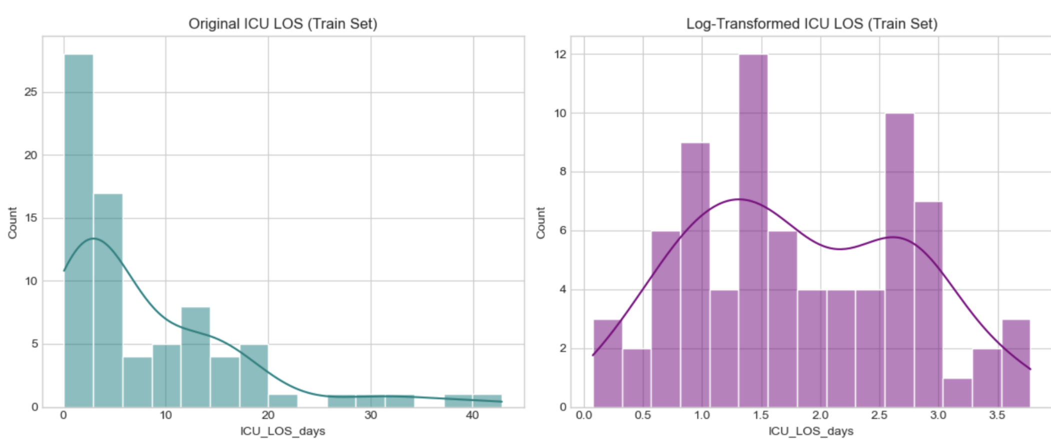


Figure 24: Distribution of ICU LOS (Days).

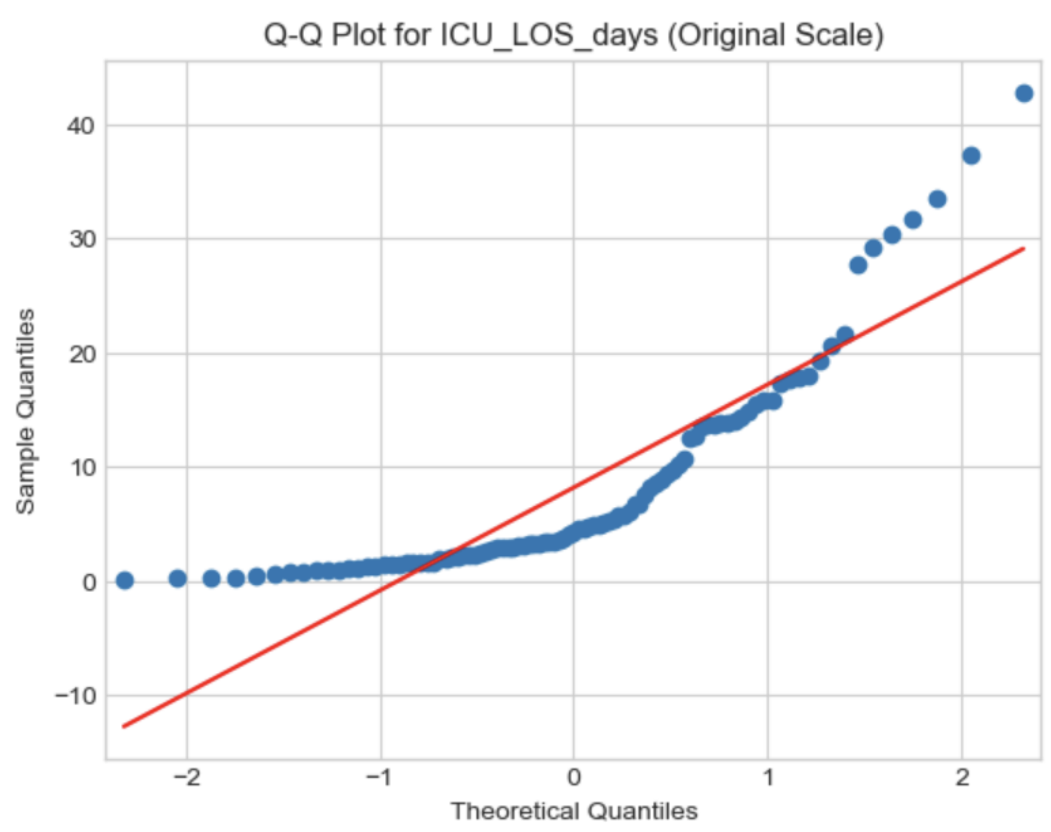


Figure 25: Q-Q Plot for ICU LOS (Original Scale).